why pressure increases as volume decreases

molecules collide with walls
more often
exert a larger force
upon same area
thus pressure increases

what happens to the supplied latent heat

thermal energy does work against intermolecular forces/breaks bonds
molecules are separated/moved apart OR becomes PE.

how evaporation cools stuff

more energetic molecules escape
less energetic molecules remain
average kinetic energy of molecules decreases
temperature depends on kinetic energy
thus temperature decreases

how a gas exerts pressure in terms of momentum

molecules collide with walls
rebound from walls
change of momentum occurs
force exerted on walls and is = total change of momentum per second
pressure exerted and is = total force/total area of walls

thermometer characteristics and what improves them

sensitivity = change in length per unit change in temperature
good sensitivity = large change in physical property(length/voltage) for small change in temperature
how to improve sensitivity = large bulb, narrow bore/capillary (thread moves further for same expansion)

range = difference b/w highest and lowest measurable temperature
how to improve range = tube sufficiently long, not too narrow bore, not too large bulb

linearity = equal size divisions/expansion for equal temperature rise same
how to ensure linearity = uniform bore, liquid expands uniformly with temperature

why can a thermocouple react quickly to rapidly changing temperatures

junction has low mass/small heat capacity;
temperature of junction reacts quickly/quickly reaches temperature of liquid;

gauze flame in low conductivity metal vs high conductivity metal

(low metal) conducts heat slowly;
above gauze:flame retains energy OR gas not hot enough to burn;
copper conducts heat rapidly;
above gauze:gas not incandescent OR gas not hot enough to burn;

latent heat def.

energy/heat required to change state/phase;
with no change in temperature;

motion of smoke particles (brownian motion)

random/haphazard/unpredictable movement;
sudden changes in direction/zig-zag motion;
appear/disappear from view;
HOW DO AIR MOLECULES CAUSE IT:
collide with smoke particles;
air molecules faster;
air molecules smaller;
move randomly;

why pressure increases with temp.

air molecules/particles hit walls;
molecules faster/higher energy when temp. raised;
greater force per unit area;

conduction in metals

at hot end molecules vibrate more
molecules collide with neighbours;
energy and vibration is passed on;

electrons transfer heat
electrons move faster/have more energy
electrons pass on energy/reach far end/free to move